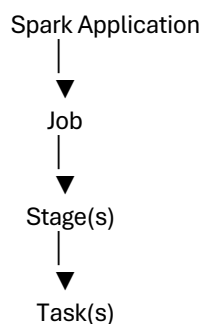


What the difference between job, stage, and task

Feature	Job	Stage	Task
Definition	The highest execution unit in Spark, representing the entire work triggered by an action.	A subset of a job containing operations that can execute without a shuffle.	The smallest execution unit that processes a single data partition.
Created When	An action is executed (show(), count(), collect(), write()).	Spark encounters a shuffle boundary (e.g., groupBy(), join(), orderBy()).	Spark creates one task for each partition in a stage.
Purpose	Executes the complete workflow needed to produce the action's result.	Organizes the job into phases separated by shuffles.	Performs the actual computation on one partition.
Granularity	Largest unit.	Medium-sized unit.	Smallest unit.
Contains	One or more stages.	One or more tasks.	No smaller execution units.
Runs On	Managed by the Driver .	Scheduled by the Driver .	Executed on Executors .
Parallelism	Jobs run independently (multiple jobs can run concurrently if resources allow).	Stages execute sequentially within a job, but tasks inside a stage run in parallel.	Tasks run in parallel across executors.
Example	df.filter(...).show() → 1 Job	filter() → Stage 1, groupBy() → Stage 2 (after shuffle).	If Stage 1 has 8 partitions , Spark creates 8 tasks .

Execution hierarchy



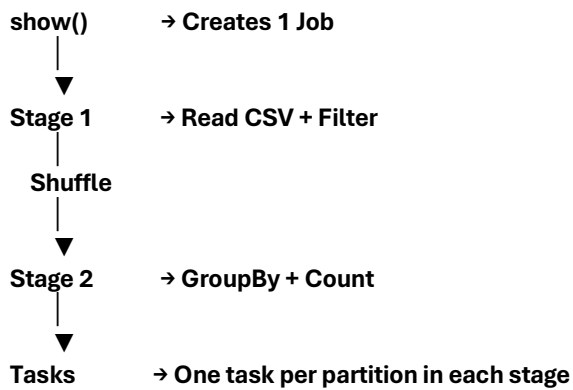
Easy way to remember

- **Job** → **What** Spark needs to accomplish (triggered by an action).
- **Stage** → **How** Spark divides the job into execution phases (separated by shuffles).
- **Task** → **Who** does the work (one task processes one partition on an executor).

Example

```
df.filter(df.salary > 5000) \  
.groupBy("department") \  
.count() \  
.show()
```

Execution flow:



Memory shortcut:

- **Action** ⇒ **Job**
- **Shuffle** ⇒ **New Stage**
- **Partition** ⇒ **Task**